

DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Program: B. Tech. Semester-III (IIOT)
Course title: Applied Linear Algebra
Course Code: MAIC-204

Time : 09:30AM-12:30PM
M.M.: 50

NOTE: Attempt all the 10 Questions. Each question carries equal marks.

1. a) Let V be the vector space of functions with basis $S = \{\sin t, \cos t, e^{3t}\}$. Let $D: V \rightarrow V$ be the differential operator defined by $D(f(t)) = d(f(t))/dt$. Compute the matrix representing D in the basis S .

- b) If w_1, w_2, w_3 are independent vectors, show that the sums $v_1 = w_2 + w_3, v_2 = w_1 + w_3, v_3 = w_1 + w_2$ is linearly independent.

2. If W is subspace of a finite dimensional vector space V , then show that every linearly independent subset of W is finite and is part of a basis.

3. a) Suppose that the linear transformation T transforms $(1, 1)$ to $(2, 2)$ and $(2, 0)$ to $(0, 0)$. Then, find $T(v)$ when $v = (2, 2)$ and $v = (3, 1)$.

- b) Let T be a linear operator on $\mathbb{R}^3$ defined by
 $(x, y, z) \mapsto (3x, -y, 2x + y + z)$.
Is T invertible? If so, find a rule for T^{-1} like the one which defines T .

4. State and prove Rank Nullity Theorem.

5. Consider the vectors

$\beta_1 = (1, 0, 1), \beta_2 = (1, 0, -1), \beta_3 = (0, 3, 4)$,
in $\mathbb{R}^3$ equipped with standard inner product space. Find out the orthonormal set of vectors $\{\alpha_1, \alpha_2, \alpha_3\}$ from these vectors using Gram Schmidt process.

6. a) Find the characteristic and minimal polynomial of the given matrix:

$$\begin{bmatrix} 4 & -2 & 2 \\ 6 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$$

- b) Find two linear operator T and U on $\mathbb{R}^2$ such that $TU=0$ and $UT \neq 0$.

7. Solve the following set of linear equation by LU-decomposition method:

$$\begin{aligned} 1x + y + z &= 1, \\ 4x + 3y - z &= 6, \\ 3x + 5y + 3z &= 4. \end{aligned}$$

8. a) Find the nature of the quadratic form:

$$x^2 - 2y^2 + z^2 + 4xy + 2yz + 6zx.$$

- b) If A is 5×5 real matrix with characteristic and minimal polynomial given below:

$$c(x) = (x-2)^3(x+7)^2 \quad m(x) = (x-2)^2(x+7).$$

what are the possible Jordan Canonical forms of A .

9. i) Consider the following polynomial in $P(t)$ with the inner product

$$\langle f, g \rangle = \int_0^1 f(t)g(t)dt:$$

$$f(t) = t + 2$$

$$g(t) = 3t - 2$$

$$h(t) = t^2 - 2t + 3$$

a) Find $\langle f, g \rangle$ and $\langle f, h \rangle$.

b) Find $\|f\|$ and $\|g\|$.

- ii) Determine whether the following function f , defined on vectors $\alpha = (x_1, x_2)$ and $\beta = (y_1, y_2)$ in $\mathbb{R}^2$ is bilinear form:

$$f(\alpha, \beta) = (x_1 - y_1)^2 + x_2 y_2.$$

10. Let V be an inner product space. Let the distance between two vectors α and β in V be defined by

$$d(\alpha, \beta) = \|\alpha - \beta\|.$$

Show that

- a) $d(\alpha, \beta) \geq 0$;
- b) $d(\alpha, \beta) = 0$ if and only if $\alpha = \beta$;
- c) $d(\alpha, \beta) = d(\beta, \alpha)$;
- d) $d(\alpha, \beta) \leq d(\alpha, \gamma) + d(\gamma, \beta)$.

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